Pedestrian flow: analytical comparison

Analytical report | 4 October 2026

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1 Pedestrian flow: what the comparison shows

Observed peak-hour counts, road-grouped OOF model predictions and existing AusRAP estimates. All results use the reviewed 49-site comparison; no model was retrained or rerun.

The along-road improvement is broad. The crossing improvement is modest and concentrated. Both average error measures favour the model, but they do not imply a uniform improvement at every site.

Table 1. Error summary of the model and the existing estimate by direction (pedestrians/hour; paired N in the first column).

Direction / paired N	Model MAE	Existing MAE	Model RMSE	Existing RMSE
Crossing / 45	15.50	18.05	29.95	36.48
Along-road / 43	21.17	52.48	38.93	74.73

MAE is mean absolute error; RMSE is the square root of mean squared error and is more sensitive to large misses. Both are pedestrians/hour; lower is better. Existing means **Existing iRAP flow estimate (recoded)**: the client's recoded and aggregated AusRAP categories, not an official precise count.

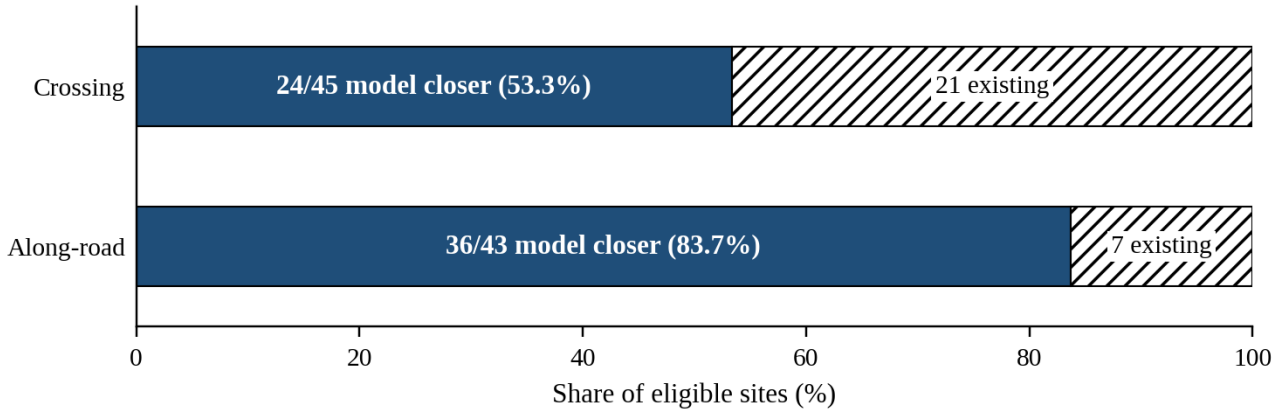


Figure 1. A site is a model win when its absolute error is smaller than the existing estimate's error. There are no ties within 1e-9. This is a descriptive comparison, not a classification accuracy or significance test.

Along-road: MAE falls from 52.48 to 21.17 (59.7% lower). The model is closer at 36/43 sites and has a median absolute error of 8.84, versus 28.33 for the existing estimate.

Crossing: MAE is 14.1% lower, but the model wins at only 24/45 sites. At an absolute-error threshold of 10 pedestrians/hour, both methods cover 29/45 sites. The two largest site improvements account for slightly more than the net crossing gain.

Remaining failure pattern: several high-flow locations are substantially underestimated by both methods. The model is not yet a reliable substitute for site review at those locations.

Coverage: all 49 sites remain in the source table. Four lack a usable existing reference; two further sites are excluded only from along-road metrics because two-side survey coverage is not established. This is internal validation, not independent external or whole-network validation.

2 Observed versus estimated flows

Shape of the relationship

Each point is one eligible site. The dashed diagonal represents exact agreement. Points below it underestimate observed flow; points above it overestimate. Both methods use identical sites and axis limits within a direction.

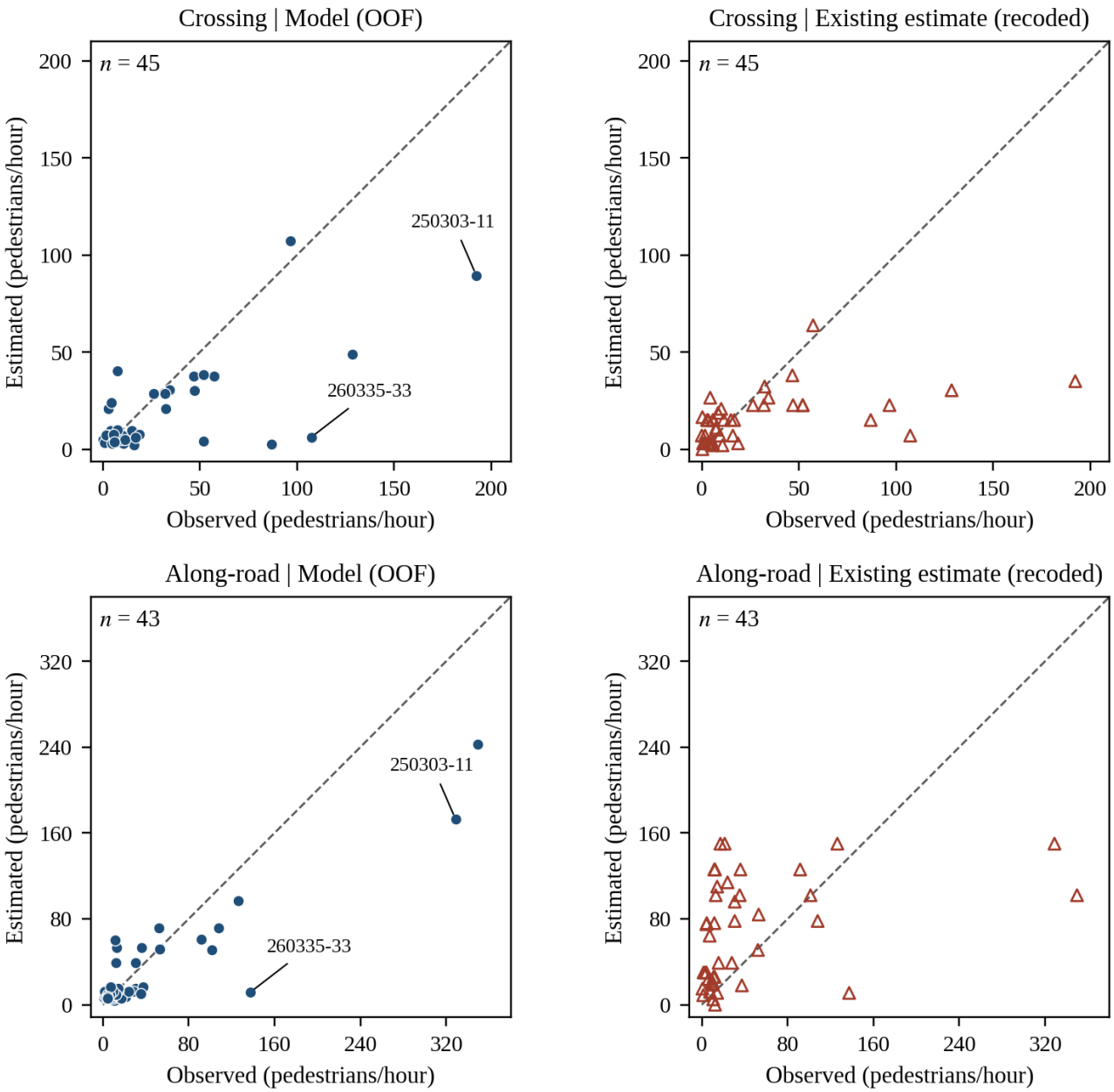


Figure 2. The values and eligible site sets are the same as the two native Excel charts. Methods are separated here to reveal overlapping low-flow points. The diagonal replaces Excel's grey equality markers; it is not a fitted regression line.

Crossing: most low-count sites cluster near the origin, while several high counts lie well above the model's estimates. Site 250303-11 has 192.33 observed pedestrians/hour and a model estimate of 89.41. That is much closer than the existing estimate of 35.00, yet it still underestimates by 102.92 pedestrians/hour.

Along-road: the existing estimates frequently sit above low observed flows. The model reduces much of that overestimation, but still misses some high-count sites. At 260335-33, the observed 137.33 contrasts with 11.59 from the model and 11.00 from the existing estimate.

3 Typical error, tail error and signed bias

Distribution, not just averages

An empirical cumulative curve answers: what fraction of sites has absolute error at or below a selected value? Higher and further left is better. No site is removed from its eligible sample.

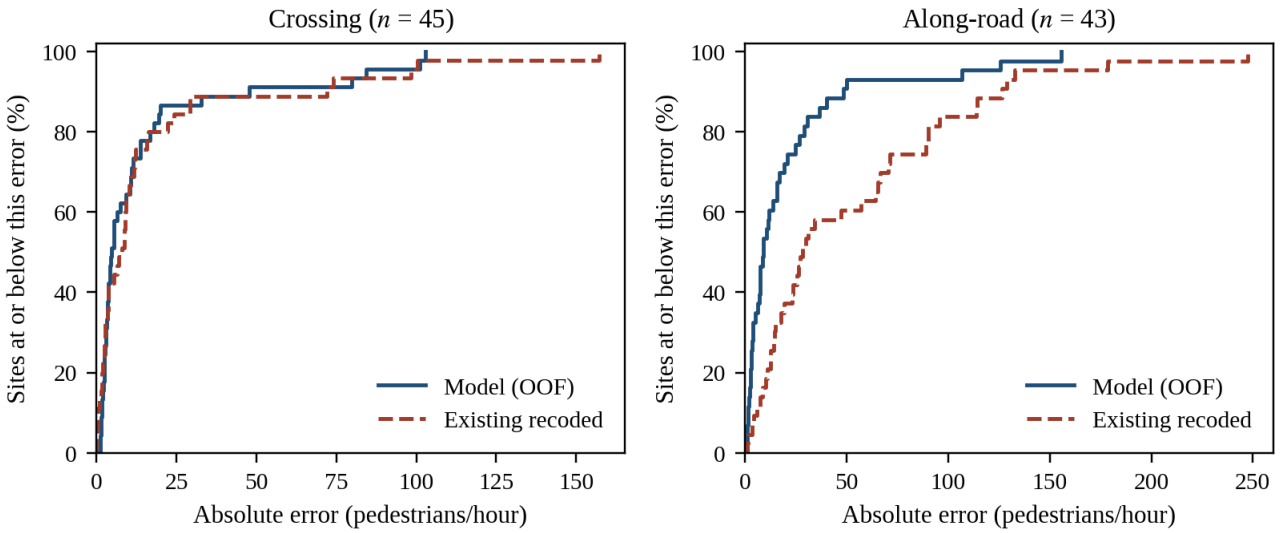


Figure 3. Cumulative absolute-error distributions. These curves retain all errors, including the largest values. Percentages describe the current sample only.

Table 2. Absolute-error (AE) and signed-error summaries by direction and method.

Direction	Method	Median AE	90th pct. AE	Mean signed error	AE ≤ 25
Crossing	Model	4.74	41.82	-9.68	39/45
Crossing	Existing recoded	7.83	54.93	-11.46	38/45
Along-road	Model	8.84	46.74	-10.69	33/43
Along-road	Existing recoded	28.33	123.93	23.42	18/43

AE = absolute error. Signed error = estimate minus observation: negative means underestimation. The 90th percentile uses linear interpolation; it is not a prediction interval.

Crossing improvement is uneven. The model's median error is lower, but both methods are within 10 pedestrians/hour at 29/45 sites. The model's 90th-percentile error remains 41.82. Mean signed errors are -9.68 for the model and -11.46 for the reference; both underestimate on average.

Along-road improvement affects much of the distribution. The model is within 25 pedestrians/hour at 33/43 sites, versus 18/43 for the reference. Reference bias is +23.42; model bias is -10.69. The model reduces the reference's overall positive bias, but leaves a negative bias concentrated among the larger observed flows.

Small errors at many sites do not cancel large absolute errors elsewhere. The gap between median error and RMSE is a reason to examine the high-error cases rather than relying on one headline score.

4 Large observed flows remain difficult

Error structure

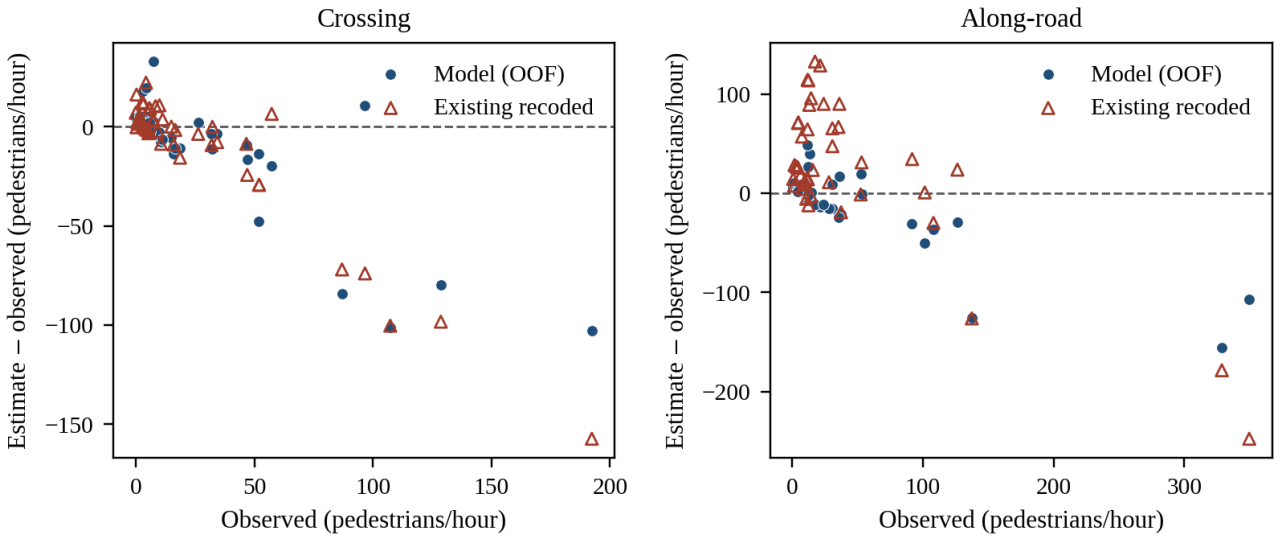


Figure 4. Signed errors against observed flow. The horizontal line is zero error. This is a diagnostic view of existing predictions; it does not establish a causal explanation.

Table 3. Errors by observed-flow group (exploratory split at 50 pedestrians/hour).

Direction	Observed group	N	Model MAE	Existing MAE	Model bias
Crossing	< 50	37	6.41	6.62	0.09
Crossing	≥ 50	8	57.52	70.89	-54.90
Along-road	< 50	34	10.45	46.55	1.68
Along-road	≥ 50	9	61.68	74.85	-57.44

The split at 50 pedestrians/hour is an exploratory descriptive threshold, not an iRAP category comparison or a pre-specified subgroup validation. Group sizes are small and depend on observed outcomes.

Crossing: among 37 sites below 50 observed pedestrians/hour, model and reference MAE are 6.41 and 6.62. At the eight sites with at least 50, they rise to 57.52 and 70.89. The model underestimates seven of those eight sites.

Along-road: below 50 observed pedestrians/hour, model MAE is 10.45 versus 46.55 for the reference. At the nine higher-flow sites, model MAE is 61.68 versus 74.85. The model underestimates eight of those nine sites. Its largest benefit is therefore among lower-flow sites with high recoded reference estimates.

The three largest model squared errors account for 69.2% of crossing squared error and 79.0% of along-road squared error. These concentrations explain why RMSE remains high even when the median error is modest.

The selected model is linear on log1p flow, but these residuals alone cannot identify whether underestimation comes from target transformation, sparse high-flow training examples, feature limitations, survey coverage or timing. Those explanations require a separate modelling and source-data investigation.

5 Which sites drive the difference?

Paired site diagnostics

Define gain as existing absolute error minus model absolute error. Positive values favour the model; negative values favour the existing estimate. Gains compare both methods against the same observation.

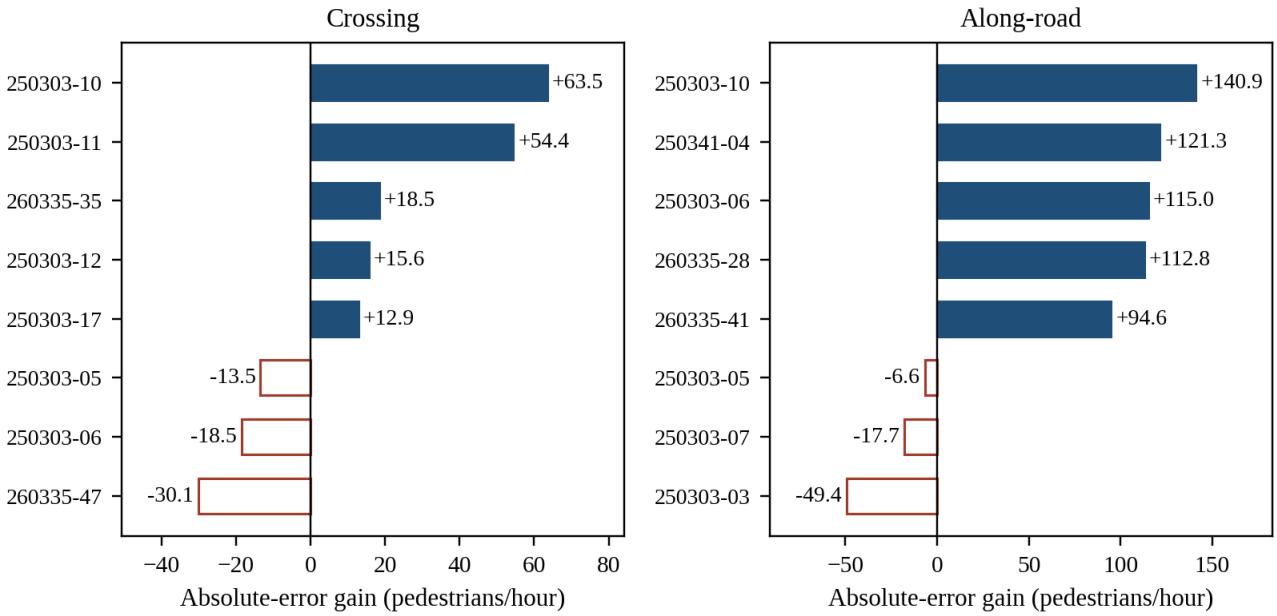


Figure 5. Five largest gains and three largest losses per direction, selected after inspecting the data. All sites still contribute to the headline metrics. The two panels use different horizontal scales.

Table 4. Selected sites illustrating the paired comparison (pedestrians/hour).

Direction	Site ID	Observed	Model	Existing	Model AE	Existing AE
Crossing	250303-10	96.67	107.14	22.67	10.47	74.00
Crossing	260335-47	7.33	40.13	10.00	32.80	2.67
Crossing	250341-04	87.00	2.71	15.00	84.29	72.00
Along-road	250303-10	349.67	242.89	102.00	106.78	247.67
Along-road	250303-03	101.33	51.29	102.00	50.04	0.67
Along-road	260335-33	137.33	11.59	11.00	125.74	126.33

Improvements and failures coexist. At 250303-10 the model is substantially closer in both directions, but its along-road count remains low. At 260335-47 it overestimates crossing, where the existing estimate is close. At 250303-03 the existing along-road estimate is almost exact, while the model misses by about 50. At 260335-33 both methods fail badly.

Site 250341-04 is already part of the outstanding corrected-target sensitivity question. This report displays its saved OOF result; it does not refit a model or remove the case from the main comparison.

6 How stable is the average advantage?

Post-hoc influence checks

The positive direction on this page means lower model MAE. These checks recompute descriptive summaries of saved predictions; they do not retrain any fold model and are not confidence intervals.

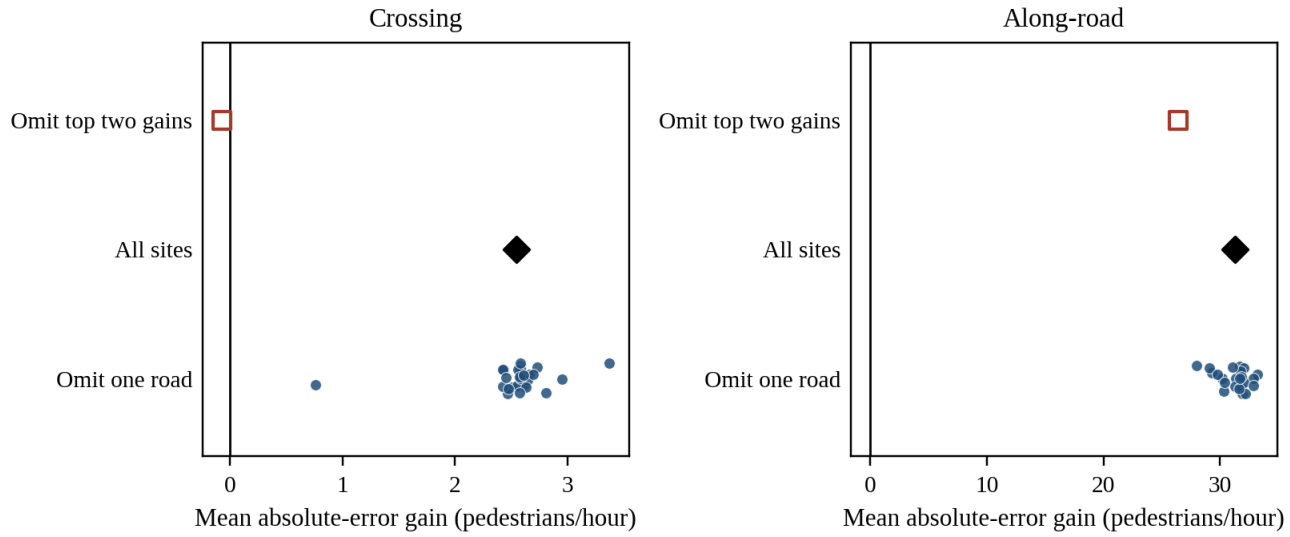


Figure 6. Blue points recompute mean gain after omitting each road in turn (26 crossing roads; 25 along-road roads). Diamonds are the full paired samples. Orange squares omit the two sites with the largest gains in each direction. Scales differ between panels.

Table 5. Influence checks on the mean absolute-error gain (pedestrians/hour).

Diagnostic: mean gain	Crossing	Along-road
All eligible sites	2.55	31.30
Leave one site out: range	1.16 to 3.29	28.70 to 33.23
Leave one road out: range	0.76 to 3.38	27.97 to 33.23
Equal weight per road	1.99	26.73
Omit two largest site gains	-0.07	26.44

Crossing is fragile to a small combination of influential cases. Sites 250303-10 and 250303-11 together contribute 117.93 pedestrians/hour of summed error reduction, exceeding the overall net reduction of 114.75. With those two omitted, the remaining 43 sites have MAE 13.58 for the model and 13.51 for the reference. This is practical parity, not evidence that the model is generally worse.

Along-road is more broadly supported in this sample. Omitting its two largest gains (250303-10 and 250341-04) still leaves an MAE advantage of 26.44 pedestrians/hour across 41 sites. Its advantage also remains positive in every one-road omission.

Omitting the largest improvements is deliberately a stress test selected using the outcomes. It must not replace the full-sample score. Equal-road weighting first averages errors within each road, then averages roads equally; it answers a different weighting question from the site-weighted main analysis.

7 What can be concluded, and where the charts come from

Interpretation and traceability

Conclusion. The saved model improves on the recoded existing estimate for along-road flow at most eligible sites. Crossing gains are smaller, with a near-even site win count and strong dependence on two favourable cases. Neither direction has solved the large underestimates at some high-flow sites. The evidence supports further validation, not an unqualified network-wide accuracy claim.

Prediction basis. OOF means out of fold. A site's road group was excluded when that fold model was trained. The workbook reuses the saved linear-regression OOF CSV; it is not a new run of the released model fitted to all 49 sites. Model and feature selection used these survey data, so the comparison is internal validation and may retain selection optimism.

How the Excel charts were generated. The workbook builder reads evidence/comparison.json, selects direction-specific eligible rows, sorts them by Site ID, and writes frozen numeric helper values on Errors. It creates native scatter charts on Summary using these ranges:

Table 6. Cell ranges on the Errors sheet used by the native Excel scatter charts.

Chart	Observed X	Model Y	Existing Y	Equality Y
Crossing, N=45	O2:O46	P2:P46	Q2:Q46	R2:R46
Along-road, N=43	T2:T44	U2:U44	V2:V44	W2:W44

All ranges above are on Errors. Each equality Y cell repeats its observed X. The PDF scatter plots redraw those same numeric pairs with a diagonal line. Other figures are newly calculated diagnostics from the same eligible records. The builder validates every helper value and the saved native chart references before producing this report. Workbook and existing explanatory report are unchanged.

Exclusions and meaning. Four sites have no usable client baseline: 250303-15, 260335-31, 260335-32 and 260335-40. Along-road additionally excludes 250341-01 (South plus Walkway, coverage unconfirmed) and 260335-38 (West only). These are not removed because of prediction error. Recoded counts are customer-defined category transformations and road-record averages; along-road references combine sides. Differences in spatial coverage, peak timing and observation period remain possible.

Priority follow-up. Review the shared large misses at 260335-33 and the large-flow targets; complete the already outstanding refit excluding corrected sites 250341-04 and 260335-38; then evaluate the frozen modelling choice on new observed roads. The two coverage exclusions and the post-hoc omission checks in this report do not substitute for that refit or external validation.

Sources and reproducibility. Counts and eligibility: evidence/site_comparison.csv. Errors: evidence/paired_errors.csv. Original metrics: evidence/metrics.csv. Input provenance and reviewed coverage: evidence/manifest.json and evidence/preflight_audit.json. These trace to the model dataset, saved OOF CSV, client merged CSV and AusRAP Words workbook. Exact fingerprints, plot source rows, derived statistics and package versions are in evidence/analysis/. Reproduce with tools/build_irap_analysis_report.py. All transformations are descriptive; no external data request or model fitting is made.

8 Appendix: every survey site

Complete observed and estimated values

Values are pedestrians/hour, rounded here to two decimals. Calculations retain full precision. Existing = client-recoded reference. Both = eligible in both directions; Cross = crossing only; Neither = reference unavailable. Available along-road values remain shown for coverage exclusions.

Table 7. Observed and estimated flows at every survey site (pedestrians/hour).

Site ID	Cross observed	Cross model	Cross existing	Along observed	Along model	Along existing	Included
250303-01	0.33	2.98	0.00	12.33	5.43	0.00	Both
250303-02	4.00	9.50	3.00	7.67	15.01	12.00	Both
250303-03	47.00	30.36	22.67	101.33	51.29	102.00	Both
250303-04	16.00	2.22	7.00	11.00	3.65	26.00	Both
250303-05	57.33	37.45	63.75	108.00	71.40	78.00	Both
250303-06	52.00	4.16	22.67	21.33	7.71	150.00	Both
250303-07	46.67	37.45	38.00	52.33	71.40	51.00	Both
250303-08	34.33	30.61	26.50	53.00	51.77	84.00	Both
250303-09	26.33	28.54	22.67	13.00	52.94	102.00	Both
250303-10	96.67	107.14	22.67	349.67	242.89	102.00	Both
250303-11	192.33	89.41	35.00	328.67	172.84	150.00	Both
250303-12	52.00	38.31	22.67	91.67	61.09	126.00	Both
250303-13	3.00	20.86	15.00	12.33	39.24	126.00	Both
250303-14	32.33	20.86	32.25	30.67	39.24	96.00	Both
250303-15	0.33	4.38	--	5.33	7.62	--	Neither
250303-16	15.00	9.55	15.00	30.67	14.86	78.00	Both
250303-17	0.33	3.61	16.50	10.00	6.01	18.67	Both
250341-01	8.33	4.25	18.67	20.33	6.32	58.00	Cross
250341-02	10.67	3.12	2.00	10.00	7.21	20.00	Both
250341-03	5.33	4.03	2.00	15.67	6.83	39.00	Both
250341-04	87.00	2.71	15.00	17.33	5.97	150.00	Both
250341-05	0.67	3.50	3.00	0.67	6.87	15.00	Both
250341-06	0.00	4.74	7.00	1.33	8.67	8.83	Both
250341-07	2.67	7.15	15.00	28.00	12.21	39.00	Both
260335-28	3.33	6.76	15.00	11.67	13.20	126.00	Both
260335-29	9.33	5.07	7.00	10.67	8.97	5.00	Both
260335-30	5.67	7.15	3.00	1.67	12.21	30.00	Both
260335-31	3.00	6.10	--	8.33	14.09	--	Neither
260335-32	2.00	7.13	--	12.33	10.52	--	Neither
260335-33	107.33	6.18	7.00	137.33	11.59	11.00	Both
260335-34	1.67	3.43	3.00	3.67	6.94	30.00	Both
260335-35	128.67	48.79	30.33	126.33	97.08	150.00	Both
260335-36	18.67	7.69	3.00	37.33	16.49	18.00	Both
260335-37	10.00	7.15	20.75	24.00	12.21	114.00	Both
260335-38	4.33	23.89	26.50	18.33	43.30	21.00	Cross

Site ID	Cross observed	Cross model	Cross existing	Along observed	Along model	Along existing	Included
260335-39	3.00	6.76	3.00	14.33	13.20	11.00	Both
260335-40	53.67	19.41	--	66.33	32.81	--	Neither
260335-41	7.33	9.77	11.00	14.33	15.40	110.00	Both
260335-42	3.33	4.91	3.00	12.00	9.57	26.00	Both
260335-43	32.00	28.54	22.67	36.00	52.94	126.00	Both
260335-44	1.00	3.43	3.00	3.00	6.94	30.00	Both
260335-45	11.33	4.91	15.00	4.67	9.57	76.00	Both
260335-46	1.67	7.15	7.00	9.67	12.21	22.00	Both
260335-47	7.33	40.13	10.00	11.67	60.11	76.00	Both
260335-48	5.67	4.55	3.00	6.33	9.33	24.00	Both
260335-49	5.67	7.69	15.00	7.33	16.49	64.33	Both
260335-50	4.67	3.12	3.00	4.33	7.21	30.00	Both
260335-51	16.67	6.00	15.00	35.33	10.57	102.00	Both
260335-52	6.00	3.61	15.00	4.33	6.01	75.00	Both

The four unavailable references are shown as --, not zero. Source observations, OOF predictions and recoded estimates are preserved exactly in the accompanying machine-readable evidence.